The Zarankiewicz number $z(13, 19; 3, 3)$ equals 120

Research note

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Abstract. The Zarankiewicz number $z(m, n; 3, 3)$ is the largest number of ones in an $m \times n$ 0/1 matrix with no 3×3 submatrix of ones. We prove $z(13, 19; 3, 3) = 120$. Matrices with 120 ones were known; the best published upper bound was 122. The proof that no such matrix has 121 ones is computer-assisted and certified throughout. A counting argument leaves 30 column-degree profiles. Twenty-six are refuted by SAT solving with DRAT proofs. The four most uniform profiles are split into 2 478 *anchor cubes* — by how the other columns meet a column of maximum degree — and the hardest cubes into 58 186 *sub-cubes* by fixing further columns in canonical order. Each sub-cube is refuted either by an exact rational Farkas certificate for a linear relaxation (32 602 of them) or by SAT with a DRAT or LRAT proof. An audit regenerated every formula, compared it byte for byte with the certified one, and re-verified every Farkas certificate in exact arithmetic. On the way we found that the LRAT checker `lrat-check` accepts an unjustified final step; we used a strict variant. All data, code and logs are public.

1 Result

The values of $z(m, n; 3, 3)$ for $m, n \le 18$ were tabulated by Collins, Riasanovsky, Wallace and Radziszowski [9], building on the classical bound of Kővári, Sós and Turán [19]; recent work raised many lower bounds and settled several values beyond that range [1–8, 10]. The cell (13, 19) was open.

> **Theorem 1.** *$z(13, 19; 3, 3) = 120$.*

Lower bound. A 13×19 matrix with 120 ones and no 3×3 block of ones is in the repository (`witness/z-13-19-120.txt`; row degrees 12, 9^{12}, column degrees 8, 7^4, 6^{14}); it was found by the search laboratory of AperioBot (§9), in which a language model writes the search programs, independently of Norris's matrix [7]. *Upper bound.* Sections 2–6 show that no admissible 13×19 matrix has 121 ones; this suffices, since deleting ones keeps a matrix admissible. Table 1 lists what was known.

Source	year	lower	upper	remark
Tan [5]	2022	—	126	SAT-assisted table
Davies, Gill, Horsley [10]	2025	—	125	LP bounds (their Table 2)
Borsik [8]	2025	119	—	BSc thesis; matrices public
Bhan, Nobili, Langer [1]	2026	114	125	their Figure 2
Afrasyab [3]; Hou [4]	2026	—	122	from $z(13, 18) = 116$, deleting a column
Saurabh [2]	2026	118	125	
Norris [7]	2026	120	122	simulated annealing
Amaro [12]	2026	120	122	found independently
Padhi [11]	2026	"120 exact"		asserted without proof; withdrawn (see below)
This note	**2026**	**120**	**120**	**certified proof**

Table 1. Bounds for $z(13, 19; 3, 3)$ in the literature (each source's own table, checked on 25 October 2026).

On [11]. This SSRN preprint lists $z(13, 19; 3) = 120$ among "44 new exact values", with no proof or data. Of its twelve exact values in rows 12–14, eight are contradicted by published matrices with more ones — for instance it gives $z(13, 20) = 124$, while a matrix with 125 ones has been public since 2025 [8] — and its $z(14, 18) = 115$ contradicts its own $z(13, 18) \ge 116$ (a row with two ones can always be added). Its Zenodo deposit was deleted as retracted. The value 120 there is therefore an assertion, not a result; we regard the proof below as the first.

2 Reduction to 30 profiles

Write r_i and c_j for row and column degrees, $\mathrm{C}(a, b)$ for binomial coefficients, and call a matrix *admissible* if it has no 3×3 block of ones, i.e. any three rows share at most two columns and any three columns share at most two rows. Suppose A is admissible, 13×19, with 121 ones.

Lemma 1 (deletion). Deleting a rows and b columns of A leaves an admissible $(13-a)\times(19-b)$ matrix, so the deleted lines carry at least $121 - z(13-a, 19-b)$ ones (a one at a crossing counted once).

We use eight upper bounds: $z(13,18) \le 116$, $z(13,17) \le 110$, $z(13,16) \le 107$, $z(12,19) \le 114$, $z(11,19) \le 106$, $z(10,19) \le 98$, $z(12,18) \le 108$ and $z(11,17) \le 96$. All follow from Table 4 of Collins, Riasanovsky, Wallace and Radziszowski [9] by the single-line case of Lemma 1 (a lightest line of an $m\times n$ matrix with Z ones has at most $\lfloor Z/n \rfloor$ ones, so $Z - \lfloor Z/n \rfloor \le z(m, n-1)$), except $z(12, 18) \le 108$, which we proved separately: under cuts from [9] alone, a 12×18 matrix with 109 ones has the single column profile $(7, 6^{17})$, refuted by Glucose 4.2.1 [16] with a 2.7 GB DRAT proof verified by drat-trim [14] (it is also claimed in [3, 4]). With it, the upper bound rests on [9] and on our certificates only.

Lemma 1 with single lines gives: every k columns of A hold at least 5, 11, 14 ones and every k rows at least 7, 15, 23 ones ($k = 1, 2, 3$). Every triple of rows lies in at most two columns, so $\Sigma_j \mathrm{C}(c_j, 3) \le 2\,\mathrm{C}(13, 3) = 572$. Exactly **30** multisets of column degrees with sum 121 satisfy these conditions (two independent enumerations agree). Each defines a case. Its formula has a variable per cell, a clause per choice of three rows and three columns forbidding nine ones, cardinality constraints fixing the column degrees and imposing the row cuts (one totalizer per row), and symmetry breaking: rows in lexicographic order, columns grouped by degree and lexicographically ordered within a degree class. This is sound because sorting rows, or swapping two adjacent equal-degree columns that violate the order, increases the matrix read row by row as a binary word, so a maximal element of every orbit satisfies all of it.

Twenty-six cases were refuted whole by Glucose 4.2.1 (`-no-pre -certified`), each DRAT proof "s VERIFIED" by drat-trim (hardest: 8 110 s of solving, 5 099 s of checking). The four most uniform profiles,

$$\mathrm{A} = (8, 7^5, 6^{13}),\ \mathrm{B} = (8, 7^6, 6^{11}, 5),\ \mathrm{C} = (7^8, 6^{10}, 5),\ \mathrm{D} = (7^7, 6^{12}),$$

did not finish in 12.5 hours. For them we strengthen the formula and split the case.

3 Mixed deletion cuts and anchor cubes

Lemma 2 (mixed cuts). For every row i and column j, $r_i + c_j - x_{ij} \ge 121 - z(12, 18) \ge 13$, and for rows i, i' and columns j, j', the four lines minus the ones at their four crossings carry at least $121 - z(11, 17) \ge 25$ ones.

This is Lemma 1 with $(a, b) = (1, 1)$ and $(2, 2)$. For instance, in profile C every row has at least 8 ones (C has a column of degree 5), and in every profile a row with 7 ones has no one in a column of degree 6. Adding these clauses made the solver 3–7 times faster on the cases below.

Anchor cubes. In the canonical form the first column has the maximum degree c and, rows being lexicographic with that column most significant, its ones fill rows $1, \ldots, c =: B$. For every other column j let $i_j = |\mathrm{col}_j \cap B|$ and $o_j = c_j - i_j$.

Lemma 3. Every admissible matrix of the case satisfies

(I) $\Sigma_j \mathrm{C}(i_j, 3) \le \mathrm{C}(c, 3)$, (III) $\Sigma_j \mathrm{C}(i_j, 2)\,o_j \le 2\,\mathrm{C}(c, 2)(13-c)$,
(II) $\Sigma_j \mathrm{C}(o_j, 3) \le 2\,\mathrm{C}(13-c, 3)$, (IV) $\Sigma_j\, i_j \mathrm{C}(o_j, 2) \le 2c\,\mathrm{C}(13-c, 2)$,

and Σi_j lies in a window fixed by the row cuts (the rows of B carry at least $c(cut_1+1) - 1$ ones when $cut_2 = 2cut_1 + 1$, and likewise the other rows).

Each sum counts row triples of a given kind (inside B, outside, two inside and one out, one in and two out) by the columns containing them; each triple lies in at most two columns, and a triple inside B already lies in the first column. A *pattern* is, for each degree class of the other columns, the multiset of their i_j; a *cube* is the case formula plus "the first column is B" plus the pattern's counts (for each class and threshold t, how many columns have $i_j \ge t$). Every canonical matrix lies in the cube of its own pattern, and all patterns meeting Lemma 3 are listed (two independent enumerations agree): A 203, B 512, C 1 254, D 509 cubes. Kissat 4.0.4 (`--unsat`, bi-

nary DRAT, drat-trim) refuted all cubes of A and B and 1 065 of C and D whole; for the other 698 the proof outgrew what drat-trim can hold in 16 GB.

4 Sub-cubes

Lemma 4 (canonical column fixing). In the canonical form, once columns 0, …, $k-1$ are fixed, the rows fall into groups with equal entries in those columns, each group consecutive and lexicographically ordered in the remaining columns. Hence column k has its ones at the start of every group and is determined by the number of its ones in each group.

So the choices for columns 1, …, d form a finite tree of count vectors; we prune it only with necessary conditions — the column's degree and the pattern's remaining counts, at most two common rows with any two fixed columns, and lexicographic order among equal-degree columns. The leaves at depth d (*sub-cubes*) cover the cube. C's 467 split cubes gave 28 845 sub-cubes (43, 15 249, 13 553 at depths 1, 2, 3); D's 231 gave 29 341 (20, 7 212, 21 756, and 353 at depth 4 under one sub-cube that took more than an hour at depth 3). A sub-cube is the cube's formula with the fixed columns as unit clauses.

5 A linear relaxation with exact Farkas certificates

We extend Afrasyab's block polytope [3, §5], which fixes one column, to all fixed columns of a sub-cube. Group the rows by their entries in the fixed columns; let n_g be the size of group g. The *type* τ of a remaining column records its number of ones in each group; let $y_{d,\tau} \geq 0$ be the number of remaining columns of degree d and type τ. Every matrix of the sub-cube gives an integral solution of

- (E) for each degree class and each value v: the columns of that class with v ones in B number exactly what the pattern says, less the fixed ones;
- (T) for each multiset t of groups (an orbit of row triples): $\Sigma\, y_{d,\tau}\, \Pi_g\, \mathrm{C}(\tau_g, t_g) \leq (2 - \varkappa_t)\, \Pi_g\, \mathrm{C}(n_g, t_g)$, where $\varkappa_t$ is the number of fixed columns containing such triples;
- (G) for each group: its rows carry at least $f \cdot n_g$ ones, where $f = \max(121 - z(12, 19), \max_j(121 - z(12, 18) - c_j))$ is the row floor of Lemmas 1–2 (7 for D, 8 for C).

If this system $Ay \leq b$, $Ey = e$, $y \geq 0$ has no real solution, Farkas' lemma gives $\alpha \geq 0$ and β with $A^{\mathrm{T}}\alpha + E^{\mathrm{T}}\beta \geq 0$ and $b^{\mathrm{T}}\alpha + e^{\mathrm{T}}\beta < 0$, and the sub-cube is empty. We find a sparse such vector with HiGHS [17] (a vertex of an ℓ_1-minimizing LP), make it rational, and verify it in exact integer arithmetic; the vector is the certificate. **32 602 sub-cubes** (C 17 359, D 15 243) were closed this way without a solver. The fixed columns are essential: with the first column alone the relaxation is invariant under the row permutations fixing it, and every anchor pattern defines a feasible point (we checked all 2 478 in exact arithmetic), so no certificate of this kind exists at cube level.

6 SAT with proofs, and a gap in an LRAT checker

The other 25 584 sub-cubes were refuted by Kissat 4.0.4 with drat-trim, or, from 23 October, by CaDiCaL 3.0.1 [13] writing an LRAT proof [15] into a pipe read by a checker while the proof is written (8 504 sub-cubes): no proof on disk, and 1.4–1.8 times faster than Kissat plus drat-trim on medium parts. A part not refuted within 25 minutes at depth 2 (60 at depth 3) was split one column deeper.

The checker. The LRAT checker `lrat-check` of the drat-trim repository (commit 2e3b2dc0ecf9) reports "c VERIFIED" for the satisfiable formula $\{x_1, x_1 \vee x_2\}$ with the one-line proof "`3 0 0`", an empty clause with no hints. A clause whose hints yield no conflict is accepted when no clause contains the negation of its first literal (a pure-literal test); for the empty clause that literal does not exist and the test succeeds. Clause identifiers are also read into 32-bit integers (an identifier of $5{\cdot}10^9$ wraps around and verifies). We found this before using the checker for any certificate, and used `lrat-check-strict` (in the repository): identical except that every added clause, the empty one included, must follow by unit propagation through its hints, and out-of-range numbers are rejected. CaDiCaL's proofs of our known-UNSAT parts verify under it, so they contain no RAT steps. Every LRAT result records the SHA-256 of both binaries, and the audit accepts LRAT results only after the checker refuses the bogus proof above.

7 Audit

An audit, separate from the runs, rechecked the whole proof from the generator (25 October 2026, log in the repository): (i) the 30 profiles are exactly those the counting admits; (ii) for each whole profile, the formula regenerated today equals the certified one byte for byte, and drat-trim re-verified every kept proof; (iii) for each anchor profile, the patterns were re-enumerated and equal the stored list; every cube's choices at its deepest depth were re-enumerated and each lies in or below a certified sub-cube; every Farkas certificate was re-verified exactly; every solver part's formula was regenerated and compared. Seventy-two proofs drawn at random were re-verified from scratch (DRAT re-checks and LRAT re-solves). No problem was found.

profile	cubes	whole	split	sub-cubes	Farkas	SAT (LRAT)	problems
26 others	—	26	—	—	—	—	0
A $(8, 7^5, 6^{13})$	203	203	0	—	—	—	0
B $(8, 7^6, 6^{11}, 5)$	512	512	0	—	—	—	0
C $(7^8, 6^{10}, 5)$	1 254	787	467	28 845	17 359	11 486 (4 081)	0
D $(7^7, 6^{12})$	509	278	231	29 341	15 243	14 098 (4 423)	0

Table 2. The certificate, by profile, as audited. About 440 hours of solving on one 8-core laptop (Apple M2, 16 GB), 17–24 October 2026.

8 Data and code

The repository github.com/Oleg-Fedorenko/zarankiewicz-13x19, archived as doi:10.5281/zenodo.23637633, contains the generator and checkers (Python, with PySAT [18]); the build recipe and hashes of the solvers and checkers, with `lrat-check-strict.c`; for every cube and sub-cube, its record (route, solver, checker, verdict, times) and the SHA-256 of its certified formula; all Farkas certificates; the audit logs; the witness; and a standalone verifier that repeats (i)–(iii) of §7 from the repository alone and re-solves random parts. The DRAT proofs of the 26 whole profiles and of the 12×18 link (3.4 GB) are in the Zenodo record; those of the cubes and sub-cubes (577 GB compressed) are kept by the author and available on request; LRAT proofs were checked while written and are not stored; any part can be regenerated and re-proved in minutes.

In addition to the main result, the repository includes explicit witness matrices establishing certified lower bounds for 18 adjacent cells (for instance $z(14, 23) \geq 146$, $z(15, 22) \geq 150$ and $z(15, 23) \geq 156$; the list is in `witness/README.md`). They were found independently and reproduce, cell by cell, the best bounds that [7, 8] obtained by heuristic search, and they exceed the bounds of [1, 2].

Consequences. With Theorem 1, Lemma 1 gives $z(14, 19; 3, 3) \leq 129$ (128 is attained [7]); the same method is being applied to 14×19.

9 Disclosure and acknowledgements

This result was obtained as a test task for AperioBot, an add-on for AI agents created by the author, a lawyer at the Aperio Bar Association; the witness matrices were found by its search laboratory.

AI assistance. This work was done with extensive help from AI systems. Claude (Anthropic) — Claude Opus 5.5 in Claude Code — wrote the encodings and programs, ran the computations and the audit, and drafted this note; independent Claude agents reviewed the code and checked the literature, and one of them found the gap in `lrat-check` described in §6. GLM proposed the anchor cubes and enumerated the patterns independently; a discussion with Claude Fable suggested the row-cut encoding. The author directed the work, decided each step and is responsible for the content. No step of the proof rests on a model's word: each rests on a matrix, a program or a certificate that anyone can check.

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